

Radar Feature Discriminability for Maritime Vessel Classification:

A Physics-Constrained Open Problem

Prepared for expert consultation — strictly confidential, not for circulation

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BACKGROUND AND PROBLEM

A land-based coastal surveillance radar collects amplitude returns from maritime vessels at ranges of 5–50 km. The goal is to classify vessel type (Fishing / Tug / Cargo / Tanker) using only the radar return — with no reliance on AIS transponder data, which can be spoofed or disabled. After exhaustive derivation and statistical evaluation of every feature obtainable from the amplitude-only radar output, it has been established that Cargo and Tanker vessels **cannot be reliably discriminated** using the current feature set.

The conclusion is that this is a measurement physics constraint, not a processing limitation. This document seeks expert guidance on whether the existing radar receiver output contains additional physical information — accessible through receiver tuning or processing parameter changes alone — that could provide the needed discriminatory power.

Hard constraint: no hardware modifications are possible. The radar operates within a secure facility. Transmitter, antenna, waveform, carrier frequency, and polarisation are fixed. Only receiver settings, clutter filter parameters, integration mode, and post-detection processing choices are adjustable.

DATASET SUMMARY

Vessel class	Detections	Tracks	Range (median)
Fishing (small commercial fishing vessel)	1,878	65	14 km
Tug (harbour / ocean tug)	1,823	37	20 km
Cargo (bulk carrier / container ship)	4,500	264	30 km
Tanker (oil / chemical / LNG tanker)	4,109	156	30 km
Total	12,310	510	

Ground truth vessel type is from AIS records matched by position and time. AIS-reported dimensions (length, beam) are excluded from all analysis: they are transponder-provided labels, not radar observables.

FEATURES DERIVED FROM THE RADAR RETURN

All features below are derived solely from the per-detection amplitude and spatial extent outputs of the existing radar processing chain. The R^4 range correction ($\ln A + 4 \ln R$) is applied to amplitude features to remove range-dependent sensitivity variation.

Feature	Physical basis
Log peak RCS	Peak cell amplitude, R^4 -corrected — vessel's dominant scatterer
Log total RCS	Sum of all cell amplitudes, R^4 -corrected — total scattering power
RCS concentration	Peak / total ratio — point vs. distributed scatterer
Aspect ratio	Cross-range extent / down-range extent — vessel orientation proxy
Footprint area (m ²)	Product of range and azimuth extents — coarse vessel size
Down-range extent (m)	Detection span in range — vessel radial length proxy
Cross-range extent (m)	Detection span in azimuth — vessel lateral size proxy
Range (m)	Slant range — retained as a confound covariate, not a vessel property

STATISTICAL EVIDENCE FOR THE FEATURE CEILING

Cohen's d (standardised mean difference) quantifies inter-class separation per feature. Values $d < 0.5$ indicate distributions that overlap substantially; $d \geq 1.0$ is the practical threshold for reliable single-feature discrimination.

Feature	Fis-Tug	Fis-Car	Fis-Tan	Tug-Car	Tug-Tan	Car-Tan
Log peak RCS	0.47	1.29	1.80	0.84	1.25	0.20
Log total RCS	0.87	1.59	2.52	0.82	1.49	0.33
RCS concentration	0.98	0.51	1.32	0.13	0.25	0.26
Aspect ratio	0.24	0.49	1.00	0.16	0.53	0.44
Footprint (m ²)	0.92	1.36	2.09	0.52	1.17	0.44
Down-range (m)	0.29	1.14	1.45	0.98	1.28	0.05
Cross-range (m)	0.90	0.98	1.15	0.13	0.38	0.56
Maximum	0.98	1.59	2.52	0.98	1.49	0.56

Every class pair except Cargo-Tanker achieves $d \geq 0.98$ on at least one feature. Cargo and Tanker reach a ceiling of $d = 0.56$ across all features combined. Both vessel types operate at identical median ranges (30 km), ruling out range bias. Cargo and Tanker are both large steel-hulled vessels of overlapping physical size; their amplitude-only radar signatures are near-indistinguishable by any combination of the features above.

EXHAUSTIVE FEATURE ENGINEERING: CEILING CONFIRMED

Three additional computational approaches were applied to verify the ceiling is not a processing artefact and that no transformation of the existing amplitude data recovers the missing discriminatory information.

(1) Temporal fractal complexity of per-track RCS sequences. Higuchi fractal dimension, Petrosian FD, sample entropy, and Hurst exponent were computed on the scan-to-scan RCS amplitude sequence for each vessel track. These quantify whether the *fluctuation pattern* across antenna rotations — rather than the mean amplitude level — differs between vessel structural types. Best Cargo-Tanker d : Hurst exponent on log-total-RCS = 0.245; Higuchi FD = 0.160. Neither approaches the $d = 0.56$ ceiling; the fluctuation pattern does not encode structural differences accessible from the existing processed data.

(2) Feature-space self-similarity (local intrinsic dimensionality). Local Intrinsic Dimensionality (LID) and multiscale neighbourhood density were computed per detection across five feature subspaces to test whether Cargo and Tanker occupy geometrically distinct regions of the feature space — independent of their marginal amplitude distributions.

Subspace	Best Car-Tan d	Feature type
Amplitude only (3-D)	0.393	Neighbourhood density
RCS-5 features (5-D)	0.333	Neighbourhood density
Without range (7-D)	0.226	Correlation dimension
Full feature set (8-D)	0.194	Neighbourhood density
Geometry only (4-D)	0.185	Local intrinsic dimension
Amplitude feature ceiling	0.560	Cross-range extent

Cargo and Tanker detections are **geometrically co-located** in every subspace tested: same neighbourhood density, same local dimensionality, same correlation dimension. Restricting to the amplitude-only subspace gives the highest self-similarity separation ($d = 0.393$), but still 30% below the direct feature ceiling. Adding geometric dimensions progressively dilutes rather than reveals structure.

Conclusion. The $d = 0.56$ ceiling is not a feature engineering gap. No nonlinear transformation, fractal measure, or geometric analysis of the existing amplitude-based features recovers additional discriminatory information. The physically missing observable must originate from a different property of the radar return — one not present in the post-integration per-scan amplitude output of the current processing chain.

THE QUESTION POSED TO THE EXPERT

Given a fixed-hardware, amplitude-only, single-polarisation marine surveillance radar: does the electromagnetic return from a large steel vessel at 20–50 km encode physical information that distinguishes a structurally complex vessel (container ship: heterogeneous deck cargo, cranes, hatches) from a structurally simpler vessel (tanker: smooth low-profile hull and deck), and if so, can this information be extracted through receiver tuning or processing parameter changes alone — without any hardware modification?

Specific aspects where expert insight is sought:

- 1. Scan-to-scan amplitude fluctuation statistics.** The vessel RCS changes from one antenna rotation to the next. A container ship’s structurally complex deck may produce a different fluctuation regime (Swerling model, variance, autocorrelation length) than a smooth tanker hull. *Is this distinction physically expected? How many consecutive scans are needed to estimate it reliably, and which receiver integration setting best preserves this information?*
- 2. Within-dwell amplitude distribution.** Multiple pulses illuminate the vessel per beam dwell. The

per-pulse amplitude sequence before integration may encode structural complexity. *Is this within-dwell distribution (skewness, pulse-to-pulse variance) expected to differ between vessel structural types? Is it accessible through receiver output logging without hardware change?*

3. **Doppler content and clutter filter tuning.** The existing clutter filter suppresses slow targets including vessel Doppler. *If the filter cutoff is lowered (a tunable parameter), is bulk vessel Doppler or sub-resolution scatterer motion (propeller, rotating machinery) expected to appear in the output? Would that motion signature differ between a diesel-direct-drive cargo vessel and a slow-speed tanker drive system, and at what filter setting would it become accessible?*
4. **Clutter interaction geometry.** The vessel modifies the surrounding sea clutter field (shadow, bow-wave, wake). This is present in the radar picture but not currently extracted. *Is this interaction geometry expected to differ between vessel types? What detection threshold or CFAR guard-band adjustment would expose it?*
5. **Any other receiver-accessible physical observable.** *Is there a property of the radar-target interaction — not listed above — that is physically expected to correlate with vessel structural type and is accessible through receiver settings or processing changes on this fixed-hardware system?*

For any property identified, the author requests an assessment of: (a) its physical origin, (b) the specific receiver or processing parameter that controls its observability, and (c) a rough expectation of the inter-class separability relative to the $d = 0.56$ ceiling currently achieved.

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